PROPOSAL FOR DISCUSSION

Histology Redesigned: executive brief

Closing handoff gaps in the histology workflow: measure first, configure first

Prepared for: Director of Operations, Platinum Dermatology
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Date: [meeting date] · **Status:** proposal for discussion

Read this first. No one has endorsed, approved, funded or bought this work: not the laboratory, ModMed or any vendor. The demo runs on my own laptop with synthetic data for a fictional lab, "Lumen Bay Dermatopathology Laboratory (fictional)", and has never been connected to EMA, the lab network or patient information. All time and effort figures are **rough estimates to refine in Phase 0**. No compliance claim is made.

1. Why I'm raising this

At the bench I keep seeing **the same specimen information copied by hand at several handoffs** (about four months here [to confirm: time at the bench as of the meeting date], plus earlier lab work elsewhere):

- Accessioners remember which office goes to which dermatopathologist.
- Parts A and B sometimes arrive as separate orders, which are hard to fix after accessioning.
- Replies to our clarification messages to offices can be delayed. We don't yet know why.
- Gross descriptions are typed into EMA after the batch, some nights until about 9 p.m.
- Many slide labels may be applied by hand on a busy day, and "the bible" is printed and color-coded by hand.

These are my observations, not measured rates. I'd like to measure them properly.

2. The idea in one sentence

A companion to EMA, not a replacement: it coordinates the handoffs between people and systems, while EMA keeps accessions, orders, gross reports and diagnoses. (The demo software is called Histology Lab OS.)

3. What the five-minute demo shows

One fictional two-part case from courier to Operations, plus two earlier fictional cases:

- A courier delivery is checked against what was expected. Offices are routed by a written, versioned rule; an unknown office is flagged, never guessed.
- Two orders for one patient: the system **suggests** they are related, a person decides, nothing is merged.
- A bottle says *left*, the order says *right*: one clarification, and **only Part B** is held until the corrected order arrives. Part A's gross record moves on.
- A tray missing a slide can't be marked complete; a printer timeout shows "unknown", not "printed"; a wrong block scan is rejected.
- Operations shows what is blocked, who owns the next step and how long it has waited.

4. Real versus simulated

- **Works today, on synthetic data, with automated tests:** the workflow rules above; the server checks each person's role on every action; a later edit to the activity record shows up when it is checked; backups are encrypted and a restore is checked before use; health checks; automated accessibility checks; and a build that is not set up shows no data. Sign-in works with a test sign-in service only. Simulated EMA messages are checked for duplicates and conflicts.
- **Simulated:** EMA and every outside connection: orders, accession numbers, office replies, printers, scanners, and the drafting "assistant".
- **Blocked:** real sign-in and production use, until the organization provides a sign-in service and hosting.
- **Needs a person to check:** screen readers, paper printing, real scanners and printers, touchscreens, and clinical validation.

Evidence (latest full run, 2026-10-24): about 800 automated tests pass (643 unit, integration and contract; 139 demo browser; 26 secured sign-in), plus a clean type check, on synthetic data. They are re-run on the final code before the meeting. **Passing tests show the code does what the tests say. They are not clinical validation.**

5. What it is not

Not validated, and not compliant because of its technical controls. Not connected to anything. Not a diagnostic tool; no AI decides anything. Not a purchase request, and not a claim that EMA or any product lacks a feature.

6. The ask: Phase 0 only

Four weeks of discovery and governance. No patient data, no software installed on lab systems, no purchase.

- **Who:** you as sponsor, the Medical Director, the IT lead, the Compliance/Privacy officer, and me.
- **Phase 0 effort (rough):** you, about 4 h over the 4 weeks plus Gate 0; the other three roles [to estimate at kickoff: a 30-minute kickoff, a 30-minute Gate 0 review, plus their discovery items]; me [to confirm: hours per week agreed with my manager, and who covers my bench].
- **What:** answer the discovery checklist; observe and time the current workflow (staff consent, no patient identifiers in notes); agree metric definitions; inventory EMA, any separate LIS [to confirm], label printers and scanners; a 30-minute small-group review of the synthetic demo; questions to ModMed and any LIS vendor about what is already licensed and configurable, sent through [to confirm: the EMA administrator or IT lead]. No contract, no data.
- **Output at week 4:** a written recommendation to **configure, buy, build, or stop**.

7. Timeline at a glance (rough estimates, weeks from approval)

Phase	Weeks	Gate
0. Discovery and governance	1–4	Configure, buy, build, or stop
1. Synthetic sandbox at the lab, no patient data	5–10	Usability; intended use approved; support model and connector developer named
2. Technical foundation and interface feasibility, test data only	8–22 (2–6 months, vendor-dependent)	Secured mode works in a **test** environment, with real sign-in and test messages
3. Validation and training (lab's quality system)	20–28	Validation approved; go-live readiness review
4. Limited pilot: one workflow, one bench, shadow first	28–36	Weekly review against the baseline
5. Decision	36–38	Continue, expand, change or stop

Best case: about 7–9 months from approval to the end of a limited pilot, if a vendor specification and test environment are available by about week 15 (about 3 months after approval); **up to about 12 months** with vendor or contract delays. **The cheapest decision-grade path is Phases 0–1 only: about 10 weeks, no patient data.** One caution: if the lab works inside EMA's own Pathology module [to confirm in Phase 0], orders may never travel to a separate system, and the build path would likely stop at Gate 0. If Phase 0 chooses configure or buy, the plan is redrawn with ModMed or the vendor.

8. What success looks like

- **The problem gets solved, even if this software is not what solves it.** Solving it with EMA configuration or an existing product is a success, and so is stopping at a gate for a written reason.
- Decisions rest on measured baselines, not my impressions. Any pilot is one workflow on one bench, measured against the baseline, including the extra checking it adds.

9. Next step

If you agree: a 30-minute Phase 0 kickoff with the Medical Director, IT lead and Compliance/Privacy officer [to confirm: date] to confirm the participants and the consent approach for observation. I'll bring the discovery checklist and a draft observation plan. *More detail: the Director FAQ and the timeline (handed out with this brief); the pilot proposal and discovery checklist, available on request.*

Beyond the pilot (vision only, not part of this ask)

If the pilot proves its value, the foundation is designed to grow, for example into sign-out and result delivery to EMA. That would be a separate, later decision needing a funded team (rough ranges in the optional *Product vision*).

Titles in italics are project documents, available on request. Highlighted [to confirm: …] *items are open questions, not facts.*